

EEE2047S: Signals and Systems I

Class Test

29 September 2022

SOLUTIONS

Please do not put your name or student number anywhere on this script.

Peoplesoft ID:

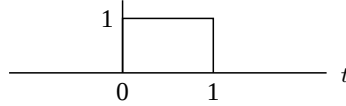
Information

- The test is closed-book.
 - This test has *four* questions, totaling 20 marks.
 - There is an information sheet attached at the end of this paper.
 - Answer *all* the questions.
 - You have 60 minutes.
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1. (5 marks) Consider the periodic signal

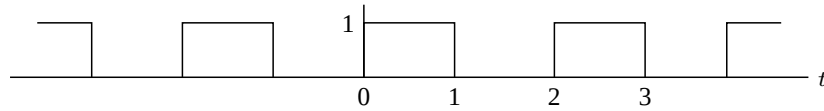
$$x(t) = \sum_{k=-\infty}^{\infty} g(t - 2k),$$

where $g(t)$ is the function below:



- (a) Sketch $x(t)$.
- (b) Find the Fourier transform of $g(t)$.
- (c) The signal $x(t)$ can be expressed as an impulse train convolved with $g(t)$. Use this to find the Fourier transform of $x(t)$.

- (a) Sketch below:



- (b) Since

$$p_1(t - 1/2) \xleftrightarrow{\mathcal{F}} \text{sinc}\left(\frac{\omega}{2\pi}\right) e^{-j\omega/2}$$

we have

$$G(\omega) = \text{sinc}\left(\frac{\omega}{2\pi}\right) e^{-j\omega/2}.$$

- (c) We can write $x(t) = p(t) * g(t)$, where

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - 2n) \xleftrightarrow{\mathcal{F}} P(\omega) = \pi \sum_{k=-\infty}^{\infty} \delta(\omega - k\pi).$$

Now

$$\begin{aligned} X(\omega) &= P(\omega)G(\omega) = \pi G(\omega) \sum_{k=-\infty}^{\infty} \delta(\omega - k\pi) = \sum_{k=-\infty}^{\infty} \pi G(k\pi) \delta(\omega - k\pi) \\ &= \sum_{k=-\infty}^{\infty} \left(\pi \text{sinc}\left(\frac{k}{2}\right) e^{-jk\pi/2} \right) \delta(\omega - k\pi). \end{aligned}$$

2. (5 marks) Find the inverse Fourier transform of

$$X(\omega) = \frac{1}{6\omega^2 - 13j\omega - 6}.$$

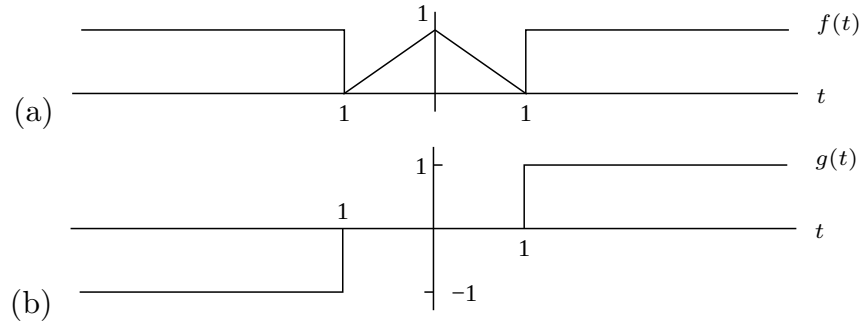
We can write

$$\begin{aligned} X(\omega) &= -\frac{1}{6} \frac{1}{-\omega^2 + 13/6j\omega + 1} = -\frac{1}{6} \frac{1}{(j\omega)^2 + 13/6(j\omega) + 1} \\ &= -\frac{1}{6} \frac{1}{(j\omega + 3/2)(j\omega + 2/3)} = \frac{a}{j\omega + 3/2} + \frac{b}{j\omega + 2/3} \end{aligned}$$

with $a = 1/5$ and $b = -1/5$. Thus

$$x(t) = \frac{1}{5}e^{-3/2t}u(t) - \frac{1}{5}e^{-2/3t}u(t).$$

3. (5 marks) Find the Fourier transforms of the functions whose graphs are shown below:



- (a) We can write

$$f(t) = (1 - p_2(t)) + (1 - |t|)p_2(t)$$

so the transform is

$$F(\omega) = 2\pi\delta(\omega) - 2\text{sinc}\left(\frac{\omega}{\pi}\right) + \text{sinc}^2\left(\frac{\omega}{2\pi}\right).$$

- (b) We can write

$$g(t) = u(t - 1) - u(-(t + 1))$$

Now $\mathcal{F}\{u(t - 1)\} = (\pi\delta(\omega) + \frac{1}{j\omega})e^{-j\omega}$ and $\mathcal{F}\{u(-(t + 1))\} = (\pi\delta(-\omega) - \frac{1}{j\omega})e^{j\omega}$, so

$$G(\omega) = \pi\delta(\omega) + \frac{1}{j\omega}e^{-j\omega} - \pi\delta(\omega) + \frac{1}{j\omega}e^{j\omega} = \frac{2}{j\omega}\cos(\omega).$$

4. (5 marks) Provide answers to the following two questions:

- (a) You are given that the Fourier transform of a Gaussian pulse $x(t) = e^{-\frac{t^2}{2}}$ is $X(\omega) = \sqrt{2\pi}e^{-\frac{\omega^2}{2}}$. Determine the Fourier transform of

$$y(t) = e^{-\frac{t^2}{2\sigma^2}},$$

where $\sigma > 0$.

- (b) You are given the Fourier transform pair below:

$$x(t) = \cos\left(\frac{\pi t}{2}\right) \text{rect}\left(\frac{t}{2}\right) \xleftrightarrow{\mathcal{F}} X(\omega) = \frac{4\pi \cos(\omega)}{\pi^2 - 4\omega^2}.$$

Determine the Fourier transform of

$$y(t) = \frac{4\pi \cos(t)}{\pi^2 - 4t^2}.$$

- (a) Observe that

$$y(t) = e^{-\frac{(t/\sigma)^2}{2}} = x(\sigma^{-1}t).$$

From the time scaling property we have

$$Y(\omega) = \frac{1}{\sigma^{-1}} X\left(\frac{\omega}{\sigma^{-1}}\right) = \sigma X(\sigma\omega) = \sqrt{2\pi}\sigma e^{-\frac{\sigma^2\omega^2}{2}}$$

- (b) Applying duality to the pair

$$\cos\left(\frac{\pi t}{2}\right) \text{rect}\left(\frac{t}{2}\right) \xleftrightarrow{\mathcal{F}} \frac{4\pi \cos(\omega)}{\pi^2 - 4\omega^2}$$

yields the new pair

$$\frac{4\pi \cos(t)}{\pi^2 - 4t^2} \xleftrightarrow{\mathcal{F}} 2\pi \cos\left(\frac{-\pi\omega}{2}\right) \text{rect}\left(\frac{-\omega}{2}\right)$$

The LHS is the quantity given so

$$Y(\omega) = 2\pi \cos\left(\frac{-\pi\omega}{2}\right) \text{rect}\left(\frac{-\omega}{2}\right) = 2\pi \cos\left(\frac{\pi\omega}{2}\right) \text{rect}\left(\frac{\omega}{2}\right).$$

INFORMATION SHEET

Fourier transform properties

Property	Transform Pair/Property
Linearity	$ax(t) + bv(t) \iff aX(\omega) + bV(\omega)$
Time shift	$x(t - c) \iff X(\omega)e^{-j\omega c}$
Time scaling	$x(at) \iff \frac{1}{a}X(\frac{\omega}{a}) \quad a > 0$
Time reversal	$x(-t) \iff X(-\omega) = \overline{X(\omega)}$
Multiplication by power of t	$t^n x(t) \iff j^n \frac{d^n}{d\omega^n} X(\omega) \quad n = 1, 2, \dots$
Frequency shift	$x(t)e^{j\omega_0 t} \iff X(\omega - \omega_0) \quad \omega_0 \text{ real}$
Multiplication by $\cos(\omega_0 t)$	$x(t)\cos(\omega_0 t) \iff \frac{1}{2}[X(\omega + \omega_0) + X(\omega - \omega_0)]$
Differentiation in time domain	$\frac{d^n}{dt^n} x(t) \iff (j\omega)^n X(\omega) \quad n = 1, 2, \dots$
Integration	$\int_{-\infty}^t x(\lambda)d\lambda \iff \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$
Convolution in time domain	$x(t) * v(t) \iff X(\omega)V(\omega)$
Multiplication in time domain	$x(t)v(t) \iff \frac{1}{2\pi}X(\omega) * V(\omega)$
Parseval's theorem	$\int_{-\infty}^{\infty} x(t)v(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \overline{X(\omega)}V(\omega)d\omega$
Parseval's theorem (special case)	$\int_{-\infty}^{\infty} x^2(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) ^2 d\omega$
Duality	$X(t) \iff 2\pi x(-\omega)$

Common Fourier Transform Pairs

$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$	$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$
1 $(-\infty < t < \infty)$	$2\pi\delta(\omega)$
$-0.5 + u(t)$	$\frac{1}{j\omega}$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$\delta(t)$	1
$\delta(t - c)$	$e^{-j\omega c} \quad (c \text{ any real number})$
$e^{-bt}u(t)$	$\frac{1}{j\omega + b} \quad (b > 0)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0) \quad (\omega_0 \text{ any real number})$
$p_{\tau}(t)$	$\tau \text{sinc} \frac{\tau\omega}{2\pi}$
$\tau \text{sinc} \frac{\tau t}{2\pi}$	$2\pi p_{\tau}(\omega)$
$\left(1 - \frac{2 t }{\tau}\right) p_{\tau}(t)$	$\frac{\tau}{2} \text{sinc}^2 \left(\frac{\tau\omega}{4\pi}\right)$
$\frac{\tau}{2} \text{sinc}^2 \frac{\tau t}{4\pi}$	$2\pi \left(1 - \frac{2 \omega }{\tau}\right) p_{\tau}(\omega)$
$\cos(\omega_0 t + \theta)$	$\pi[e^{-j\theta}\delta(\omega + \omega_0) + e^{j\theta}\delta(\omega - \omega_0)]$
$\sin(\omega_0 t + \theta)$	$j\pi[e^{-j\theta}\delta(\omega + \omega_0) - e^{j\theta}\delta(\omega - \omega_0)]$
$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\frac{2\pi}{T})$

with $p_{\tau}(t) = u(t + \tau/2) - u(t - \tau/2)$ and $\text{sinc}(\lambda) = \sin(\pi\lambda)/(\pi\lambda)$.

Laplace transform properties

Property	Transform Pair/Property
Linearity	$ax(t) + bv(t) \iff aX(s) + bV(s)$
Time shift	$x(t-a)u(t-a) \iff e^{-as}X(s) \quad a \geq 0$
Time scaling	$x(at) \iff \frac{1}{a}X(\frac{s}{a}) \quad a > 0$
Frequency differentiation	$t^n x(t) \iff (-1)^n X^{(n)}(s)$
Frequency shift	$e^{at}x(t) \iff X(s-a)$
Differentiation	$x'(t) \iff sX(s) - x(0^-)$ $x''(t) \iff s^2X(s) - sx(0^-) - x'(0^-)$ $x^{(n)}(t) \iff s^nX(s) - s^{n-1}x(0^-) - \dots - x^{(n-1)}(0^-)$
Integration	$\int_{0^-}^t x(\lambda)d\lambda \iff \frac{1}{s}X(s)$ $\int_{-\infty}^t x(\lambda)d\lambda \iff \frac{1}{s}X(s) + \frac{1}{s} \int_{-\infty}^0 x(\lambda)d\lambda$
Time convolution	$x(t) * v(t) \iff X(s)V(s)$
Frequency convolution	$x(t)v(t) \iff \frac{1}{2\pi j}X(s) * V(s)$

Initial value: $f(0^+) = \lim_{s \rightarrow \infty} sF(s)$

Final value: $f(\infty) = \lim_{s \rightarrow 0} sF(s)$ with all poles in left-hand plane

Common Unilateral Laplace Transform Pairs

$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s)e^{st}ds$	$X(s) = \int_0^\infty x(t)e^{-st}dt$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{\lambda t}u(t)$	$\frac{1}{s-\lambda}$
$te^{\lambda t}u(t)$	$\frac{1}{(s-\lambda)^2}$
$t^n e^{\lambda t}u(t)$	$\frac{n!}{(s-\lambda)^{n+1}}$
$\cos(bt)u(t)$	$\frac{s}{s^2+b^2}$
$\sin(bt)u(t)$	$\frac{b}{s^2+b^2}$
$e^{-at} \cos(bt)u(t)$	$\frac{s+a}{(s+a)^2+b^2}$
$e^{-at} \sin(bt)u(t)$	$\frac{b}{(s+a)^2+b^2}$
$re^{-at} \cos(bt+\theta)u(t)$	$\frac{(r \cos \theta)s + (ar \cos \theta - br \sin \theta)}{s^2 + 2as + (a^2 + b^2)}$
$re^{-at} \cos(bt+\theta)u(t)$	$\frac{0.5re^{j\theta}}{s+a-jb} + \frac{0.5re^{-j\theta}}{s+a+jb}$
$re^{-at} \cos(bt+\theta)u(t)$	$\frac{As+B}{s^2+2as+c}$ $r = \sqrt{\frac{A^2c+B^2-2ABa}{c-a^2}}, b = \sqrt{c-a^2}, \theta = \tan^{-1} \frac{Aa-b}{A\sqrt{c-a^2}}$

Trigonometric identities

$$\begin{aligned} \sin(-\theta) &= -\sin(\theta) & \cos(-\theta) &= \cos(\theta) & \tan(-\theta) &= -\tan(\theta) & \sin^2(\theta) + \cos^2(\theta) &= 1 \\ \sin(2\theta) &= 2\sin(\theta)\cos(\theta) & \cos(2\theta) &= \cos^2(\theta) - \sin^2(\theta) = 2\cos^2(\theta) - 1 = 1 - 2\sin^2(\theta) \\ \sin(\theta_1 + \theta_2) &= \sin(\theta_1)\cos(\theta_2) + \cos(\theta_1)\sin(\theta_2) & \cos(\theta_1 + \theta_2) &= \cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2) \\ e^{j\theta} &= \cos(\theta) + j\sin(\theta) \end{aligned}$$